



COEP Technological University, Pune

A Unitary Public University of Government of Maharashtra

(formerly College of Engineering Pune)

Wellesley Road, Chhatrapati Shivajinagar, Pune - 411005.

Department of Applied Sciences and Humanities

Mathematics Section

Course: Numerical Methods (NM)(All except Planning)

Course Code	OEC-24015	Scheme of Evaluation	
Teaching Plan	2+0+0+1=2	TA	20
Credits	2	MSE	30
		ESE	50

Syllabus: Numerical Solution of Algebraic and Transcendental equations: Bisection Method, Secant Method, Regula-Falsi Method, Newton-Raphson Method, Successive Approximation Method, Applications. Solution of linear simultaneous equations: Homogeneous/Non-homogeneous systems, Gauss Elimination, Gauss Jordan, Gauss-Seidel Methods, LU-Decomposition, Cholesky Method, Applications.

Interpolation: Langrange's Interpolation, Newton's forward, backward and central difference method, divided difference method, Inverse Interpolation, Applications. Numerical Differentiation: Forward, Backward and Central Difference Methods, Applications.

Numerical integration: Trapezoidal Rule, Simpson 1/3rd and 3/8th Rule, Gauss Quadrature - Two and Three Point Formula, Applications.

Curve Fitting: least square criteria- 1st and 2nd Degree, Applications.

Numerical Solution of Ordinary Differential Equation: Taylor Series Method, Euler Method, Modified Euler Method, Runge Kutta 2nd and 4th order method, Simultaneous Differential Equations and Second Order Differential Equations, Applications. **[28 Hrs]**

Text Book :

- Chapra, S.C. & Canal, R. P., Numerical Methods for Engineers, 5th Ed., Tata McGraw Hill Publication.

Reference Books :

- Balagurusamy, E., Numerical Methods, Tata McGraw Hill Publication.
- Rajaraman, V., Computer Oriented Numerical Methods, Prentice Hall of India Ltd.
- Sastry, S. S., Introductory Methods of Numerical Analysis, Prentice Hall of India Ltd.
- Jain, M.K., Iyengar, S.R.K. and Jain, R.K., Numerical Methods for Scientific and Engineering Computations, 5th Ed., New Age International Ltd.

- Rajasekaran, S., Numerical Methods in Science and Engineering – A practical Approach, S. Chand and Co. Ltd.
 - Rao, S.S., Optimization Theory and Applications, New Age International Ltd.
 - Computed Oriented Numerical Methods, (5th edition) by R.S. Salaria, Khanna Publishing Company Private Limited, New Delhi.
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Outcomes : Students will be able to

1. know and recall core knowledge of the syllabus. (To measure this outcome, questions may be of the type- define, identify, state, match, list, name etc.)
2. understand basic concepts. (To measure this outcome, questions may be of the type- explain, describe, illustrate, evaluate, give examples, compute etc.)
3. analyze the problem and apply the appropriate concept. (To measure this outcome, questions will be based on applications of core concepts)
4. give reasoning. (To measure this outcome, questions may be of the type- true/false with justification, theoretical fill in the blanks, theoretical problems, prove implications or corollaries of theorems, etc.)
5. apply core concepts to new situations. (To measure this outcome, some questions will be based on self-study topics and also comprehension of unseen passages.)